

History & Evolution of Computers and Artificial Intelligence

To understand how the evolution of computing and AI shaped modern business innovation and decision-making.

FROM MACHINES TO MINDS

AI & BEYOND

The Evolution of Computers & AI



MILESTONES IN COMPUTING HISTORY



1642 – Passal's Mechanical Calculator (first device for arithmitic)

1837 – Charles Babbage's Analytical **Engine** (concept of programmable computer)

1940s – ENIAC & UNIVAC: first electronic general-purpose computers

1950s – Transistors replace vacuum tubes > faster, smaller, cheaper machines

1990s–2000s Internet & network-connetigns world

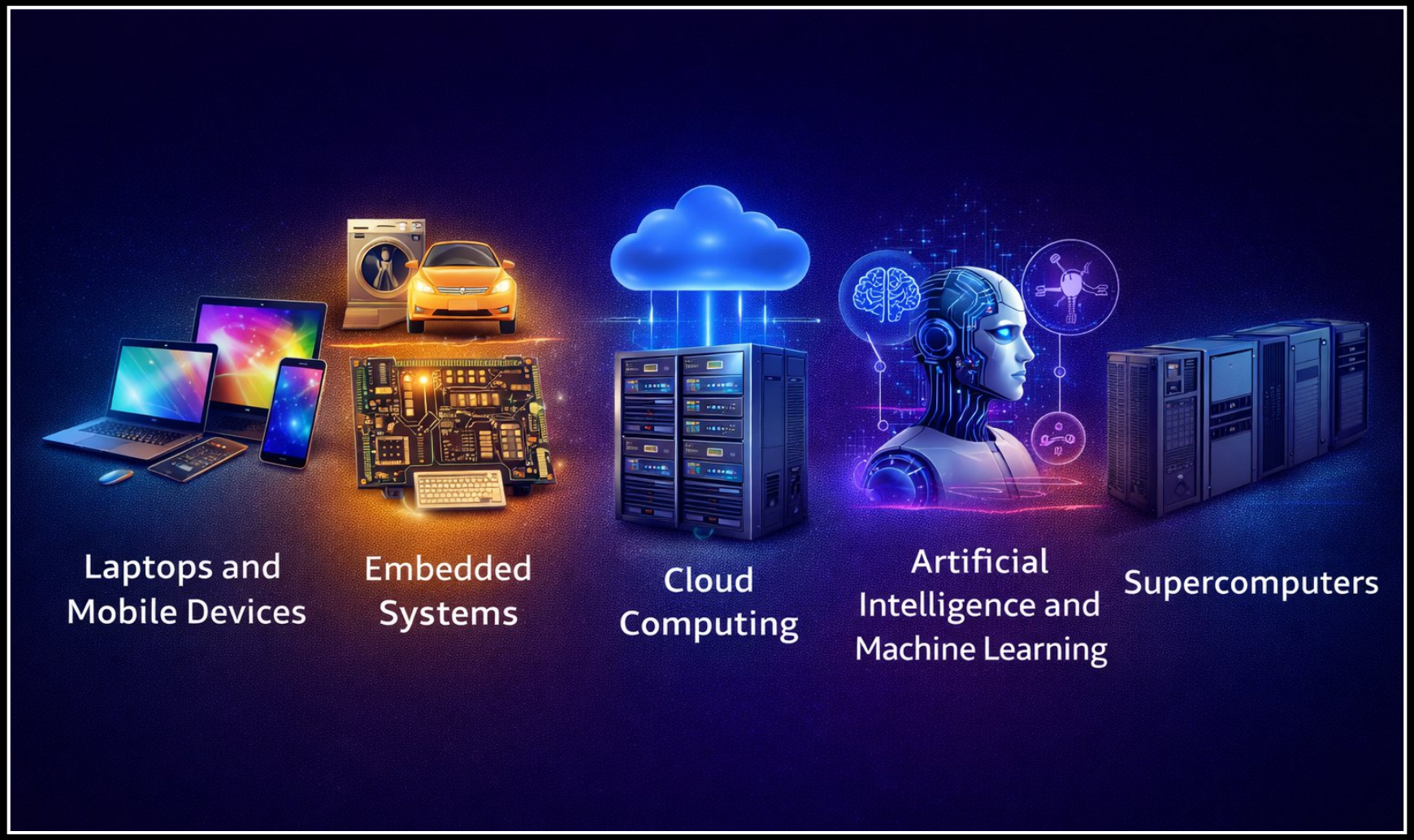
2010s–Present Cloud computing, quantum research, and AI integration

Each techn… …s process data

From Calculators to Supercomputers

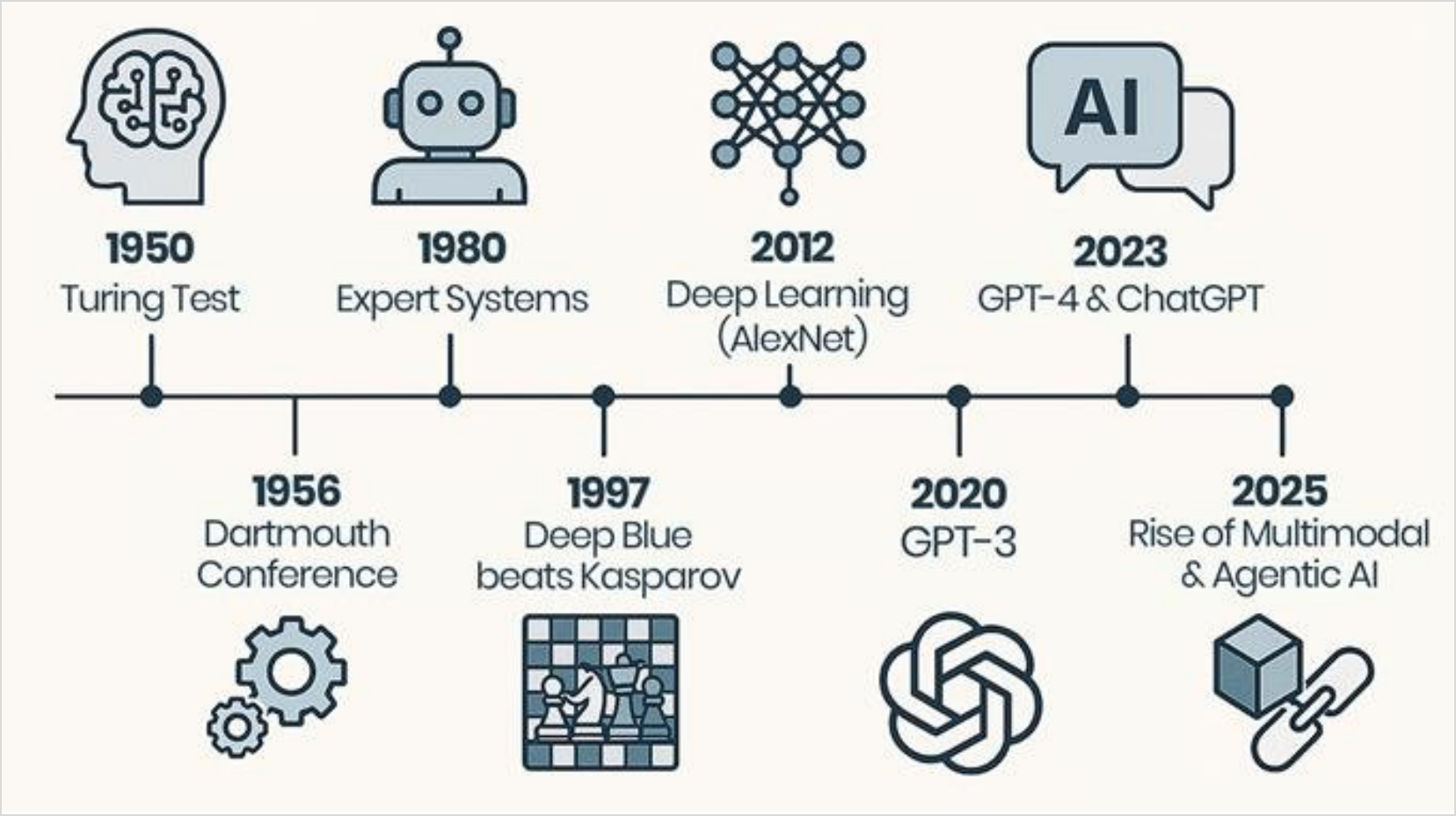


From Calculators to Supercomputers



From Calculators to Supercomputers

Evolution of Artificial Intelligence



AI Winters

It refers to a time when interest in artificial intelligence decreases, often following a period of intense hype and investment.

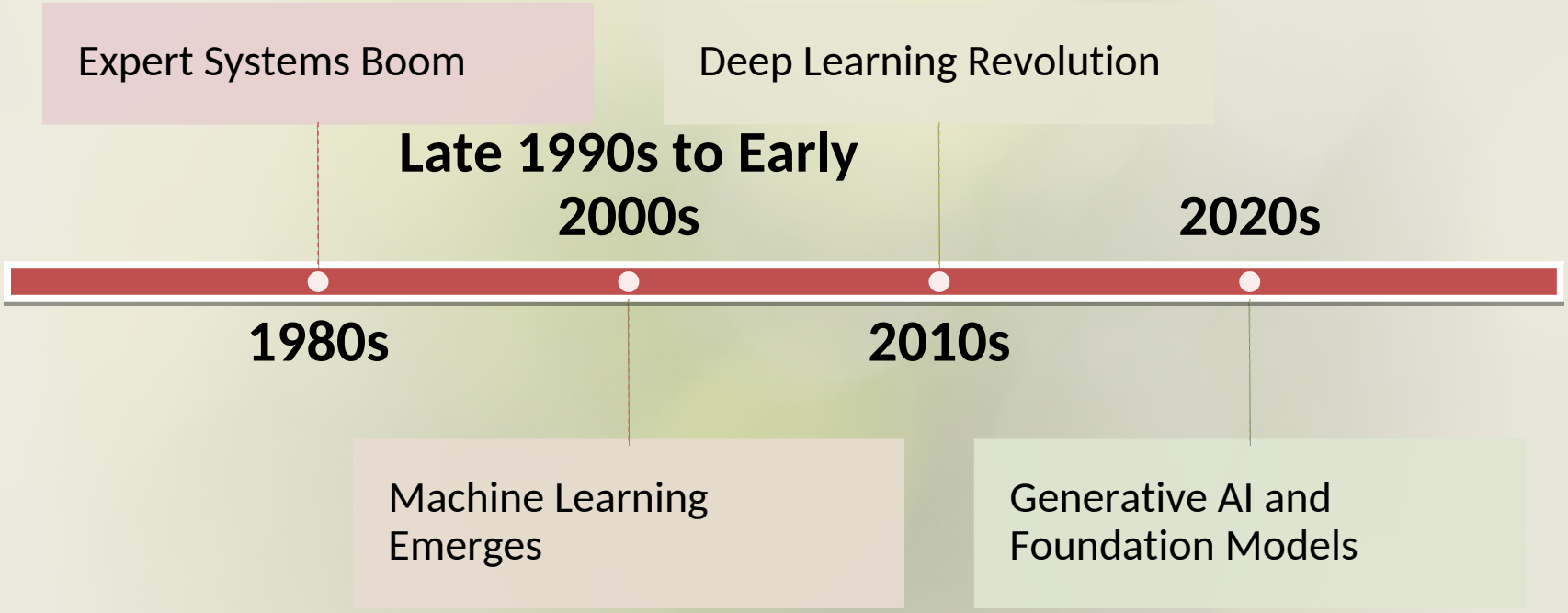
These recessions were marked by:

- **Reduced funding** for AI research
- **Public and academic doubt**
- **Slowed technological progress**
- **Fewer commercial applications**

Major Reasons were:

- Overpromising results that can't be delivered
- Technological bottlenecks, such as limited computing power or poor algorithms
- Lack of practical applications or commercial viability
- Disillusionment among investors and researchers

AI Resurgences



From expert systems in the 1980s to IBM's Deep Blue defeating chess champion Kasparov in 1997

Watson's Jeopardy! victory in 2011 and the rise of deep neural networks have since transformed industries, enabling machines to learn, reason, and solve complex problems at scale.

AI RESURGENCES– 1980s: EXPERT SYSTEMS BOOM



THE RETURN OF AI – KNOWLEDGE BECOMES POWER

WHAT ARE EXPERT SYSTEMS?
AI programs that mimic the decision-making ability of human experts in specific domains.

HOW THEY WORK
They capture expert knowledge in rules and use inference engines to solve complex problems.

IMPACT & ADOPTION
Widely used in medicine, finance, engineering, and business to improve accuracy and efficiency.

WHY IT MATTERED
Reignited AI research, attracted investments, and laid the foundation for future breakthroughs.

KEY EXAMPLES
- **MYCIN** – Medical diagnosis
- **DENDRAL** – Chemical analysis
- **XCON (R1)** – Configuring computer systems

THE 1980s JOURNEY

Early 1980s – AI focuses on rule-based systems and knowledge representation.

Mid-1980s – Expert systems gain popularity and industry adoption.

Late 1980s – Commercial boom – investments, tools, and growth.

By 1990 – Lay the groundwork for the next wave of AI innovations.

> “THE 1980s PROVED THAT **KNOWLEDGE + RULES = INTELLIGENT DECISIONS**”

AI RESURGENCES –

LATE 1990s TO EARLY 2000s:

MACHINE LEARNING EMERGES

AI

A NEW ERA FOR AI BEGINS



BEYOND RULES

From logic-based AI to data-driven learning



MACHINE LEARNING

Algorithms learn patterns from data



BREAKTHROUGHS

SVMs, Decision Trees, Bayesian Networks & more



REAL-WORLD IMPACT

Spam filters, Recommender systems, Web search & more

DATA DRIVES.
ALGORITHMS LEARN.
AI EVOLVES.

1997
IBM Deep Blue beats Kasparov

1998–2001
Growth of ML algorithms

2001
AI goes mainstream in industry & web

Early 2000s
Data + Compute Power fuel AI revival

MORE DATA • BETTER ALGORITHMS • STRONGER COMPUTING = SMARTER AI

AI Resurgences-2010s Deep Learning Revolution

- Fueled by **big data**, **GPU acceleration** and **neural networks**.
- Breakthroughs in image recognition, NLP and game playing.
- AI became central to tech giants like Google, Microsoft and Amazon.

AI Resurgences-2020s Generative AI and Foundation Models

Tools like **ChatGPT**, **DALL·E**, and **Copilot** showcase large language models and generative capabilities.

AI is now used in creative fields, coding, education, and customer service.

The rise of **multimodal models** (handling text, image, audio) marks a new frontier.

Computing Power: The Engine Behind AI



It revolutionized AI by enabling parallel processing of large datasets. This made training deep neural networks feasible in days instead of months.

Moore's Law and Beyond: While Moore's Law predicted doubling of transistor density every two years, AI's compute demand has outpaced it, growing more than twice as fast

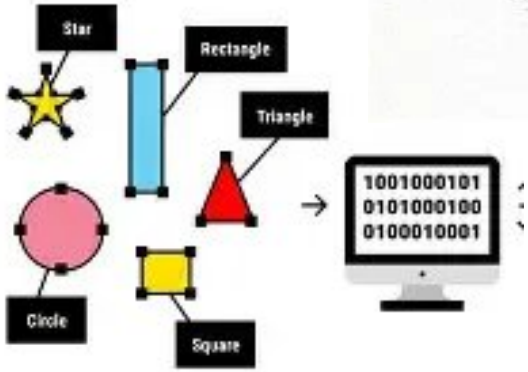


Platforms like AWS, Azure, and Google Cloud democratized access to high-performance computing, allowing startups and researchers to build powerful models without owning expensive hardware.

Data: The Fuel for Learning

- Big Data Explosion
- Labeled Datasets
- Real-Time Streams
- Synthetic Data









The Business Connection

1 Customer Experience Enhancement

- **AI chatbots** provide 24/7 support
- **Sentiment analysis** helps companies understand customer emotions.
- **Personalized recommendations** boost engagement and sales.

2 Operational Efficiency

- **Process automation** using AI-powered bots.
- **Predictive maintenance** in manufacturing reduces downtime by forecasting equipment failures.
- **Inventory optimization** ensures stock levels match demand, cutting costs.

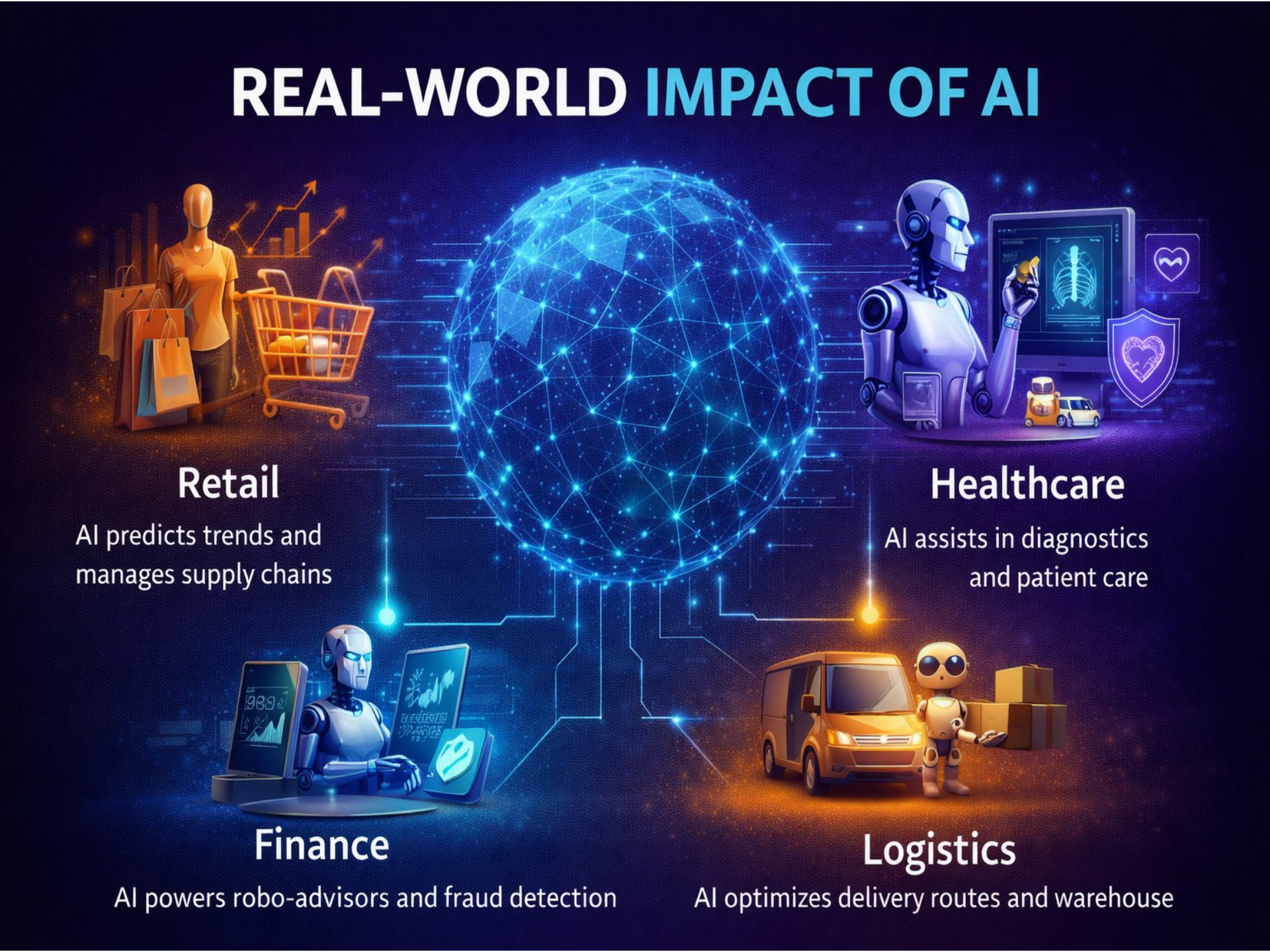
3 Marketing & Sales

- **AI-driven analytics** identify customer segments and buying patterns.
- **Lead scoring** helps sales teams prioritize prospects most likely to convert.
- **Content generation tools** assist marketers in creating blogs, ads, and social media posts faster.

4 Human Resource & Talent Management

- **Resume screening** tools identify top candidates quickly.
- **Employee sentiment tracking** helps HR address concerns proactively.
- **AI-based training platforms** personalize learning paths for employees.

REAL-WORLD IMPACT OF AI



Retail

AI predicts trends and manages supply chains

Healthcare

AI assists in diagnostics and patient care

Finance

AI powers robo-advisors and fraud detection

Logistics

AI optimizes delivery routes and warehouse

Future of AI in Business

Generative AI

- Create Tasks
- Product Design

Multi Model AI

- Combine text & image
- Also combine Voice inputs

Decision Intelligence

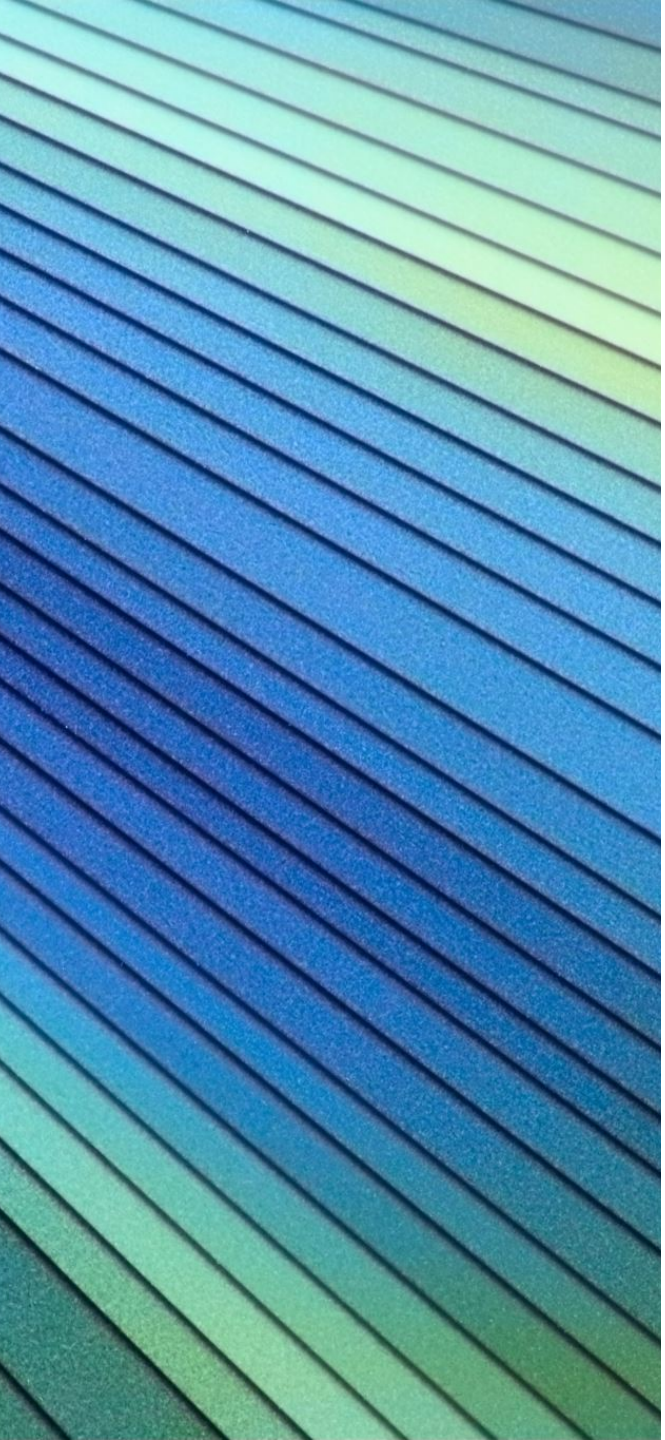
- Support Strategic planning
- Cause & Effect relationship



AI powered pricing strategy

AI-driven pricing in e-commerce is all about

- Precision
- Speed
- Personalization



AI powered pricing strategy

- Dynamic Pricing Algorithm
- Personalized Pricing
- Predictive Modelling
- Competitive Intelligence

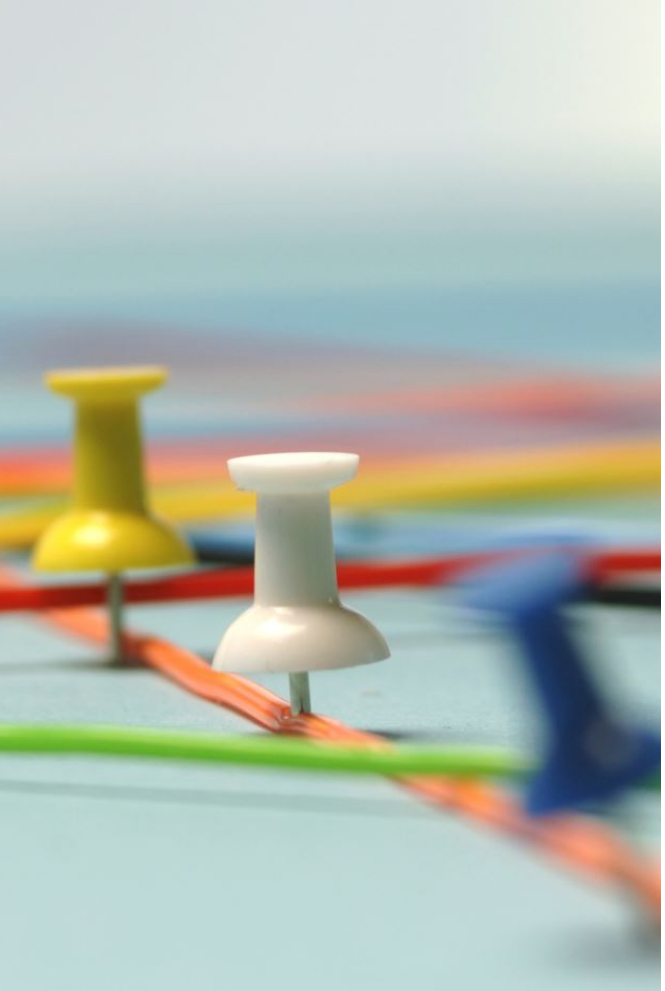


AI powered pricing strategy

A fashion retailer uses AI to:

- **Lower prices on slow-moving inventory**
- **Raise prices on trending items with high demand**
- **Offer personalized discounts based on browsing history**

- This strategy led to a 15% increase in revenue and 20% reduction in unsold stock



Activity – AI Timeline Challenge

Activity:

Create a timeline showing major AI milestones and their business impact.

Example format:

- 1956 – AI coined → Academic foundation for automation
- 1997 – Deep Blue defeats Kasparov → Rise of algorithmic strategy in finance
- 2020s – Generative AI → Business process automation & creativity boost

Outcome:

Visualize how each technological shift influenced modern business strategies.